Interview Protocol

[Introduction] During software evolution, the *implemented* architecture tends to exhibit increasing and accumulating violations of the *intended* architecture. This phenomenon is defined as **architecture erosion**, and can manifest in a variety of **symptoms**. One of the ways to detect violation symptoms of architecture erosion is by analyzing textual artifacts that contain information related to system architecture and its design. The violation symptoms can occur at different stages of development and be self-admitted (the developers admit they made the violations themselves) or pointed out by other developers. However, manual identification of violation symptoms for reporting erosion-related issues can be tedious, time-consuming, and error-prone. Therefore, we developed several machine learning and deep learning classifiers to automatically identify violation symptoms from code review comments, which can further provide warnings for potential erosion-related issues during the evolution.

[Open-ended Questions]: The open-ended questions of this interview are based on RQ3 in our study, that is, *Whether practitioners find automatic identification of violation symptoms useful in practice?*
Rationale: If the results generated by the classifiers based on ML and DL algorithms (in RQ1) are perceived useful, the automatic identification of violation symptoms from textual artifacts would be helpful in practice. This RQ aims at validating whether the results from the classifiers are useful and serve the aforementioned purpose. We plan to conduct interviews to collect more opinions on assessing the effectiveness and practicability of the trained classifiers.

[Participants]: Our collected data from the Gerrit tool contains the developers' information (i.e., names and email addresses). We will invite participants to be interviewed after reading the survey, by sending emails to the developers whose comments are identified by our classifiers.

[Data Analysis]: We will apply descriptive statistics to analyze the responses to the demographic and Likert scale questions. For the open-ended question, we will use open coding and Constant Comparison method to analyze and categorize the qualitative response.

Interview Questions

Open-ended **Interview Questions (IQs)** on the application of automatic identification of violation symptoms of architecture erosion from code review comments:

IQ1: How do you think the integration of an automated system for identifying violation symptoms of architecture erosion from code review comments would fit within the code review workflow? This system would analyze reviewer comments in real-time to flag potential or overlooked violation symptoms. By doing so, it would prevent the incorporation of code changes that violate system architecture, thereby contributing to a more robust and reliable development process. What are your thoughts on the potential benefits and challenges of such integration?
(Free text)

IQ2: Based on your experience, what strategies would you recommend to effectively use automated identification of violation symptoms in the code review process? How can this technology be utilized to maximize its benefits and efficiency in identifying and addressing architectural issues during code review?
(Free text)

IQ3: In addition to code review scenarios, can you envision other contexts or scenarios where our approach for automated identification of violation symptoms could be beneficial in software development practice? What potential applications or integrations do you see as valuable in enhancing code quality and development workflows?
(Free text)